

A Block-Diagonal Subcase of the Formal-Identity Approximation Question

FA/Banach run packet for arXiv:1108.6026

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Status

This packet proves a partial result for the question following Corollary 6.7 of Kaminska–Popov–Spinu–Tcaciuc–Troitsky [1]. The full question for arbitrary strictly singular operators $\ell_p \rightarrow d_{w,p}$, $p > 1$, remains open here.

Source Question

Let $j : \ell_p \rightarrow d_{w,p}$ be the formal identity. After proving that $\mathcal{SS}(\ell_1, d_{w,1})$ is contained in the closure of operators factoring through j , the source asks whether the same assertion remains valid for $p > 1$.

The question appears on PDF page 22:

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It follows that the operator $\tilde{S} : \ell_1 \rightarrow \ell_1$ defined by $\tilde{S}f_n = Sf_n$ belongs to $L(\ell_1)$. Obviously, $S = j\tilde{S}$. So, $\|T - j\tilde{S}\| < \varepsilon$. $\square$

The next corollary follows immediately from Theorem 6.6 and Corollary 6.4. This corollary can be considered as a support for Conjecture 6.1.

Corollary 6.7. *$\mathcal{SS}(\ell_1, d_{w,1})$ is contained in the closure of $\{jS : S \in L(\ell_1, d_{w,1})\}$.*

Question. Does Corollary 6.7 remain valid for $p > 1$?

The following fact is standard, we include its proof for convenience of the reader.

Proposition 6.8. *If X is a Banach space then $\mathcal{SS}(X, \ell_1) = \mathcal{K}(X, \ell_1)$.*

Proof. Let $T \notin \mathcal{K}(X, \ell_1)$. Pick a bounded sequence (x_n) in X such that (Tx_n) has no convergent subsequences. By Schur's theorem, (Tx_n) and, therefore, (x_n) have no weakly Cauchy subsequences. Applying Rosenthal's ℓ_1 -theorem twice, we find a subsequence (x_{n_k}) such that (x_{n_k}) and (Tx_{n_k}) are both equivalent to the unit vector basis of ℓ_1 . It follows that T is not strictly singular. $\square$

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Proof Intuition

The source proves the key column condition: if $T : \ell_p \rightarrow d_{w,p}$ is strictly singular, then the columns (Tf_n) are almost lengthwise bounded. For arbitrary operators this is not by itself enough, because truncating columns does not automatically give a bounded ℓ_p -valued operator. In the block-diagonal case, however, the supports are disjoint. The truncated columns are uniformly N -sparse and disjoint, so they define a bounded operator on ℓ_p . The discarded tails are also disjoint and uniformly small in $d_{w,p}$, so the Lorentz upper p -estimate makes the tail operator small in norm.

Definitions

Let $1 < p < \infty$, and let $w = (w_n)$ be a Lorentz weight: $w_1 = 1$, $w_n \downarrow 0$, $w_n > 0$, and $\sum_n w_n = \infty$. For $x \in c_0$, write x^* for the non-increasing rearrangement of $(|x_n|)$, and

$$\|x\|_{d_{w,p}}^p = \sum_{n=1}^{\infty} w_n (x_n^*)^p.$$

We say that $T : \ell_p \rightarrow d_{w,p}$ is block diagonal if, for the unit vector basis (f_n) of ℓ_p , the vectors $y_n = Tf_n$ have pairwise disjoint successive supports in the unit vector basis of $d_{w,p}$.

Lemma 1 (Disjoint upper p -estimate). *If (x_n) are pairwise disjointly supported vectors in $d_{w,p}$, then for all finitely supported scalar sequences (a_n) ,*

$$\left\| \sum_n a_n x_n \right\|_{d_{w,p}}^p \leq \sum_n |a_n|^p \|x_n\|_{d_{w,p}}^p.$$

Proof. It is enough to consider non-negative finitely supported vectors. Rearrange the entries of each x_n in decreasing order, and compare them with the global decreasing rearrangement of the disjoint union $\sum_n a_n x_n$. If the m -th largest entry of a fixed block appears at global position k , then $k \geq m$, hence $w_k \leq w_m$. Summing over all entries gives the stated inequality. $\square$

Partial Theorem

Theorem 1. *Let $1 < p < \infty$. Let $T : \ell_p \rightarrow d_{w,p}$ be a strictly singular block-diagonal operator. Then*

$$T \in \overline{\{jS : S \in L(\ell_p)\}}^{\|\cdot\|_{L(\ell_p, d_{w,p})}},$$

where $j : \ell_p \rightarrow d_{w,p}$ is the formal identity.

Proof. Put $y_n = Tf_n$. By Corollary 6.4 of [1], the sequence (y_n) is almost lengthwise bounded. Thus, for each $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that after removing the N largest coordinates of each y_n , the remaining vector has $d_{w,p}$ -norm at most ε .

For each n , let z_n be the vector obtained from y_n by keeping the N largest coordinates and setting all other coordinates to zero. Since T is block diagonal, the supports of the z_n are disjoint. Also, z_n is N -sparse and

$$\|z_n\|_{d_{w,p}} \leq \|y_n\|_{d_{w,p}} \leq \|T\|.$$

For an N -sparse vector z one has

$$\|z\|_{d_{w,p}}^p \geq w_N \|z\|_{\ell_p}^p,$$

because the first N Lorentz weights are all at least w_N . Hence $\sup_n \|z_n\|_{\ell_p} \leq w_N^{-1/p} \|T\|$.

Define $S_N : \ell_p \rightarrow \ell_p$ by $S_N f_n = z_n$. Since the supports of the z_n are disjoint,

$$\|S_N(a_n)\|_{\ell_p}^p = \sum_n |a_n|^p \|z_n\|_{\ell_p}^p \leq w_N^{-1} \|T\|^p \sum_n |a_n|^p,$$

so $S_N \in L(\ell_p)$.

Now set $r_n = y_n - z_n$. The supports of the r_n are still disjoint, and $\|r_n\|_{d_{w,p}} \leq \varepsilon$ for every n . By Lemma 1, for every finitely supported $(a_n) \in \ell_p$,

$$\|(T - jS_N)(a_n)\|_{d_{w,p}}^p = \left\| \sum_n a_n r_n \right\|_{d_{w,p}}^p \leq \varepsilon^p \sum_n |a_n|^p.$$

Thus $\|T - jS_N\|_{L(\ell_p, d_{w,p})} \leq \varepsilon$. Since $\varepsilon > 0$ was arbitrary, T belongs to the stated norm closure. $\square$

Scope And Novelty Check

This proves the source question only for block-diagonal operators. The general case remains open here because almost lengthwise boundedness of the columns does not, for $p > 1$, automatically make the column truncation map bounded from ℓ_p to ℓ_p .

Local run indexes were searched for arXiv:1108.6026 and for keywords around Lorentz sequence spaces, norm-closed operator ideals, formal identity, strictly singular operators, and ℓ_p factorization. A web search on 2026-07-03 found the source paper but no exact later answer to the $p > 1$ question. The result should be reviewed as a natural partial subcase, not as a full resolution of Conjecture 6.1 or the full question after Corollary 6.7.

References

- [1] Anna Kaminska, Alexey I. Popov, Eugeniu Spinu, Adi Tcaciuc, and Vladimir G. Troitsky, *Norm closed operator ideals in Lorentz sequence spaces*, arXiv:1108.6026.